Portland State
UNIVERSITY

Case Study: What's Driving Revenue?

The Scenario: A generous benefactor has donated an office furniture and supply business to PSU. So far, the only updates to the business have been a name change and a new tagline: Viking Supply Co: Outfitting the Halls of Valhalla and PSU. That tagline might need some workshopping. But the real issue is improving sales.The Dean of the School of Business has asked students to evaluate the company's recent performance and provide data-backed recommendations. Your team has been given one full year of sales data. Your job is to determine which product categories and regions are performing well, and where PSU might focus future improvements or investments.

After speaking with the sales leadership team, you've learned It's been a tough year for Viking Supply Co. While sales remain strong overall, leadership is seeking clarity on what's working and what's not. Some regions have faced real challenges, most notably, a hurricane earlier this year impacted operations and staffing in Florida. On a positive note, a major mobile carrier launched an aggressive marketing campaign in California and New York, offering generous trade-in incentives that may have drawn foot traffic to nearby Viking locations.

As a student intern, your job is to help the Dean make sense of the sales data and identify both obstacles and opportunities. A limited budget has been set aside, to either invest in marketing or additional sales support, but not both. Where should the money go? What deserves attention, and what should be deprioritized?

Assignment Overview

In this case study, you and your partner will work together to uncover patterns and/or key insights in the sales data. You'll clean and prepare the dataset, create pivot tables and charts, calculate summary statistics, and write a brief summary of your findings.

You'll practice using your spreadsheet of choice to:

- Clean and structure raw data for analysis
- Apply formatting and design best practices
- Use formulas like SUM, AVERAGE, and IF
- Create pivot tables and visualizations
- Make business recommendations using data

Part 2: Spreadsheet Setup and Data Preparation

The spreadsheet you're about to open won't look too different (maybe even better) than something you might be handed in a real workplace. Some of the data will be helpful for your analysis, and some won't.

Your first task is to prepare the spreadsheet: make a copy of the original data, apply consistent formatting, and follow spreadsheet best practices to ensure it's clean, organized, and easy to read.

The steps below draw from both the Excel textbook and the Excel Best Practices guide posted in Canvas. Use them as a reference for formatting, layout, and readability.

1. Open the Excel file titled **Viking Supply Co. | 2024 Sales Data.**
2. Create a copy of the existing tab. Rename the original tab **Raw Data**. Name the new tab **Clean Data**. You will work in the Clean Data tab for the rest of this assignment.
3. Clean and format your spreadsheet:
    - Bold the header row and freeze the top row
    - Adjust column widths so all content is visible, or optionally update the field labels to make them clearer and more intuitive
    - Use Wrap Text if needed, and make sure row height adjusts as needed
    - Format `Sales` and `Profit` as accounting, and `Discount` as a %
    - Round currency values to two decimals
    - Look to see if comma formatting is needed anywhere, for easier scanning of large numbers
    - Remove any blank rows or columns at the edges of the dataset
    - Add a **custom fill color** to the header row using PSU branding:
        - Primary green: `#6d8d24`
    - Delete any columns you know won't be needed for this analysis, such as **Ship Mode**, **Country**, and **Postal Code**

NOTE: When it comes to deleting columns or removing unnecessary data, not all decisions will be clear cut. In real-world projects, it's common to keep some data even if you're not using it right away, because once a report is shared, new questions often come up. In the spirit of future data exploration, keep fields like **Customer Segment** and **State.**

Part 3: Data Exploration

Now that your dataset is cleaned and formatted, it's time to explore what the data actually reveals. In this step, you'll use basic analysis tools to identify standout product categories and overall sales performance. Your goal is to uncover patterns that might help guide future business decisions.

This is also where your earlier spreadsheet skills come into play. Use functions and features you've already practiced to summarize, compare, and interpret the data.

What to Look for in Your Pivot Tables

Each of the next steps will guide you through building a pivot table to explore a different dimension of the data, by category, region, and a category-region breakdown. As you complete each pivot table, consider the following:

- Which product categories or regions generate the most revenue?
- Are there any large performance gaps between items in the same group?
- Do any categories or regions appear to be underperforming?
- What patterns or surprises stand out across these views?

NOTE: The goal isn't to arrive at a single correct answer but to help you prepare for the final recommendations.

Step 1: Analyze by Product Category

Start by examining how much revenue each product category generates. You can use a **pivot table** to organize this data.

- Click anywhere inside your cleaned dataset (no need to select the full range), then insert a pivot table. The tool will automatically detect the data range.
- Set **Product Category** as the row label
- Set **Sales** as the values field (summed, not counted)
- Be sure to apply basic formatting, like adjusting column widths
- Try sorting your pivot table by sales (highest to lowest) to see which category performs best.
    - If sorting doesn't seem to "stick" the first time, don't worry. There's more than one way to sort!
- If they aren't formatted already, format the sales totals with accounting and commas for clarity

Step 2: Add a Simple Chart to Visualize Your Findings

- Click anywhere into your pivot table and create a chart (Column).
    - Insert → Pivot Table
- Give your chart a clear, descriptive title
- If needed, move or resize your chart so it doesn't cover the data or take up too much space on the sheet.
- Rescale your chart to show clearer differences
    - If the **Chart Editor** isn't visible, **double-click** the chart and the editor should appear. Near the top, select the **Customize** tab.
    - Because we're adjusting the **Y-axis** (the vertical one showing sales totals), click on **Vertical axis** in the Customize menu.
    - You'll see options to set the **Min** and **Max** values. Try the following:
        - **Minimum**: 600,000
        - **Maximum**: 900,000

These settings should make the differences between product categories much easier to see.

Step 3: Analyze Sales by Region

- Create a second pivot table on a new sheet
- Use **Region** as the row label and **Sales** as the value
- Sort the pivot table from **highest to lowest** sales
- Add a chart to visualize the results and adjust the scale
    - Hint: Which region is throwing off the scale? Since you're only comparing regions, choose a **maximum value** that fits the highest regional sales
    - Don't forget to adjust the min value. Try a few options and observe the effect
- Don't forget basic formatting

Step 4: Analyze Sales by Region and Subcategory
This layout introduces a few more layers, but don't overthink it. You're still just dragging fields into rows, columns, and values. Same process, new perspective.

- Create a third pivot table on a new sheet
- Use Category and Sub-Category as the rows
- Use Region as the columns
- Use Sales as the value
- Format the table to make it readable, including consistent number formatting and adjusted column widths
- Look across the regions to identify which categories are strongest in each area
- Pay attention to sub-categories that underperform or overperform depending on the region
- Don't worry about adding a chart for this one. This pivot table is meant to support your final recommendations

Step 5: Organize Your Tabs

- **Rename** your pivot table sheets (tabs) to match the content:
  - `Sales by Category`
  - `Sales by Region`
  - `Category by Region`
- **Reorder** your worksheet tabs so they appear in the following order: `Sales by Category`, `Sales by Region`, and `Category by Region`.

Part 4: Summary Tab and Key Metrics

Now that you've explored sales by category and region, create a summary tab to highlight key performance metrics. This is a chance to practice basic formulas and build a simple dashboard-style view of your analysis.

Step 1: Create a New Tab

- Add a new sheet and name it **Summary**
- Use this space to show key figures using formulas like `SUM`, `AVERAGE`, and `MAX`

Step 2: Include the Following Metrics

- Total Sales (use `SUM`)
- Average Sales per Order (use `AVERAGE`)
- Highest Single Order Amount (use `MAX`)
- Total Quantity Sold
- Average Sales per Order
- Average Profit per Order
- Profit per Unit Sold (Hint: Total Profit ÷ Total Quantity)

Step 3: Add a Title and Light Styling

- Add a heading at the top of the sheet: **Key Metrics Overview**
- Use borders, spacing, or PSU brand colors to make the layout easy to read

Part 5: Business Analysis

Insert a final worksheet named **Reflection** and respond to the following prompts (2–4 sentences each). Use specific data and observations from your pivot tables or summary sheet.

1. **How do external factors like weather events or other companies' promotions affect business strategy?**
    a. Identify one example from the scenario and explain how it may have influenced Viking's regional sales performance.
2. **Which region shows the greatest potential for growth? Which might need extra support?**
    a. Consider all three pivot tables and the narrative context.
3. **Where should the business invest its limited budget? Marketing or sales support (training, tools, or staffing)?**
    a. Support your recommendation with at least one data point from your analysis and one consideration from the case scenario.
    b. As part of your recommendation, discuss whether it would be more ethical and effective to replace underperforming staff or invest in developing their skills. You're not expected to solve a real-world HR dilemma, just explore the factors a decision-maker might consider and form a point of view.
4. **What additional data would you request or explore before finalizing a recommendation?**
5. **What was most challenging about interpreting the data?**

Part 6: Final Polish

Before submitting, take a few minutes to review your file as a whole. A polished spreadsheet not only helps others understand your work, it makes you look more professional.

- Fix any visible errors, like `######` in cells, misaligned columns, or inconsistent fonts
- Check that your charts have clear titles and readable axis labels
- Make sure all currency and number formats are applied consistently
- Add a touch of branding by applying PSU colors (hex codes provided) to header rows or chart accents:
    - **Primary green**: `#6d8d24`
    - **Secondary green**: `#cfd82d`
    - **Black**: `#000000`
- Use **Primary green** (`#6d8d24`) in your header rows as a subtle branding element, for polish. If the dark green makes it difficult to read, change the font color to a high-contrast option like white.
- Look for ways to use complementary colors in your pivot table titles, chart borders, or to highlight key rows in your manual summary